RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 6 (Summary and Tutorial)

Topic: Vectors 1 (Vector Algebra, Ratio Theorem, Scalar and Vector Product)

Summary for Vectors 1

Vector Algebra

With reference to an origin $O(0, 0, 0)$, given points $A(a_1, a_2, a_3)$ and $B(b_1, b_2, b_3)$, we have the corresponding (position vectors) expressed in column form $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$.

$$\mathrm{a} \pm \mathrm{b} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \pm \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_1 \pm b_1 \\ a_2 \pm b_2 \\ a_3 \pm b_3 \end{pmatrix} \text{ and}$$

$$k\mathrm{a} = k \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} ka_1 \\ ka_2 \\ ka_3 \end{pmatrix} \text{ with } k \text{ a real number.}$$

The **magnitude** (or modulus) of a vector, $|\mathbf{a}|$, is the non-negative number

$|\mathbf{a}| = \left\| \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \right\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$. This value is equal to the distance from O to A.

We say that $\mathbf{a}$ is parallel to $\mathbf{b}$, denoted by $\mathbf{a} \parallel \mathbf{b}$, if and only if $\mathbf{b} = \lambda\mathbf{a}$ for some $\lambda \in \mathbb{R} \backslash \{0\}$, that is, $\mathbf{b}$ is a (non-zero) scalar multiple of $\mathbf{a}$.

- If $\lambda > 0$, then $\lambda\mathbf{a}$ and $\mathbf{a}$ are in the **same** direction.
- If $\lambda < 0$, then $\lambda\mathbf{a}$ and $\mathbf{a}$ are in **opposite** directions.

Points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, relative to the origin O, such that $\mathbf{b} = \lambda\mathbf{a}$ for some $\lambda \in \mathbb{R} \backslash \{0\}$. We then say that the points O, A and B are **collinear**.

The **unit vector** in the **direction** of **a** denoted by $\hat{\mathbf{a}}$ is obtained by scaling **a** by $\frac{1}{|\mathbf{a}|}$, thus $\hat{\mathbf{a}} = \frac{1}{|\mathbf{a}|}\mathbf{a}$.

The vectors $\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ -4 \\ 4 \end{pmatrix}$ are parallel since $\begin{pmatrix} -2 \\ -4 \\ 4 \end{pmatrix}$ is a scalar multiple of $\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$ $(k = -2)$ but are in opposite directions since $k < 0$. The points $(-2,-4,4)$, $(0,0,0)$ and $(1,2,-2)$ are also said to be collinear. The magnitude of $\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$ is $\sqrt{1^2 + 2^2 + (-2)^2} = 3$ so the unit vector in the direction of $\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$ is $\frac{1}{3}\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$

Let **a** and **b** be non-zero and **non-parallel** vectors:

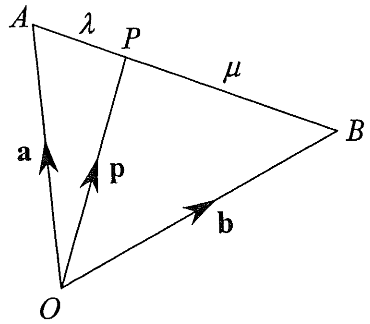
- If $\lambda\mathbf{a} = \mu\mathbf{b}$ for some $\lambda, \mu \in \mathbb{R}$, then $\lambda = \mu = 0$.
- If $\alpha\mathbf{a} + \beta\mathbf{b} = s\mathbf{a} + t\mathbf{b}$ for some $\alpha, \beta, s, t \in \mathbb{R}$, then $\alpha = s$, $\beta = t$.

Note the importance of **non-parallel** vectors when comparing coefficients.
Suppose $\mathbf{a} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$ then $6\mathbf{a} + 2\mathbf{b} = 4\mathbf{a} + 3\mathbf{b}$ however we cannot "compare coefficients" of vectors **a** and **b** (note that **a** is parallel to **b**) as $6 \neq 4$ and $2 \neq 3$.

Ratio Theorem

Consider a triangle OAB with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.
So $\mathbf{a}$ and $\mathbf{b}$ are non-zero and **non-parallel** vectors.
Let P be a point which divides AB in the ratio $\lambda : \mu$, i.e. $\dfrac{AP}{PB} = \dfrac{\lambda}{\mu}$.

If $\overrightarrow{OP} = \mathbf{p}$, then $\mathbf{p} = \dfrac{\mu\mathbf{a} + \lambda\mathbf{b}}{\lambda + \mu}$ (MF27)



Note that the Ratio Theorem is an immediate consequence of the addition of vectors.
From diagram, $\mathbf{p} - \mathbf{a} = \dfrac{\lambda}{\lambda + \mu}(\mathbf{b} - \mathbf{a})$, rearranging we have $\mathbf{p} = \dfrac{\mu\mathbf{a} + \lambda\mathbf{b}}{\lambda + \mu}$. Sometimes, it is easier to use $\mathbf{p} - \mathbf{a} = \dfrac{\lambda}{\lambda + \mu}(\mathbf{b} - \mathbf{a})$ like the following example.

Points A, B and P have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{p}$ respectively, relative to the origin O.
Given that $\mathbf{a} = \begin{pmatrix} 2 \\ 2 \\ 5 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 6 \\ -3 \end{pmatrix}$, find $\mathbf{p}$ if P lies on AB produced such that $\dfrac{AB}{AP} = \dfrac{2}{5}$.

Solution: Easier to find directly, $\mathbf{p} - \mathbf{a} = \dfrac{5}{2}(\mathbf{b} - \mathbf{a})$

[DO NOT WRITE $\dfrac{\mathbf{b} - \mathbf{a}}{\mathbf{p} - \mathbf{a}} = \dfrac{2}{5}$. We cannot divide vectors]

$$\mathbf{p} = \begin{pmatrix} 2 \\ 2 \\ 5 \end{pmatrix} + \frac{5}{2}\begin{pmatrix} 2-2 \\ 6-2 \\ -3-5 \end{pmatrix} = \begin{pmatrix} 2 \\ 12 \\ -15 \end{pmatrix}$$

Scalar (Dot) Product

Points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, relative to the origin O.

$\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$ where $\theta = \angle AOB$.

Re-arranging, $\cos\theta = \dfrac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$.

- If vectors $\mathbf{a}$ and $\mathbf{b}$ are in the **same direction** then $\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}||\mathbf{b}|$, largest possible value.
- If vectors $\mathbf{a}$ and $\mathbf{b}$ are in the **opposite direction** then $\mathbf{a}\cdot\mathbf{b} = -|\mathbf{a}||\mathbf{b}|$, smallest possible value.

Two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular, denoted by $\mathbf{a}\perp\mathbf{b}$, if and only if $\mathbf{a}\cdot\mathbf{b}=0$.
In particular, $\mathbf{i}\cdot\mathbf{j}=\mathbf{j}\cdot\mathbf{k}=\mathbf{k}\cdot\mathbf{i}=0$ as $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are mutually perpendicular.

If $\mathbf{a}=\begin{pmatrix}a_1\\a_2\\a_3\end{pmatrix}$ and $\mathbf{b}=\begin{pmatrix}b_1\\b_2\\b_3\end{pmatrix}$, then $\mathbf{a}\cdot\mathbf{b}=\begin{pmatrix}a_1\\a_2\\a_3\end{pmatrix}\cdot\begin{pmatrix}b_1\\b_2\\b_3\end{pmatrix}=a_1b_1+a_2b_2+a_3b_3$.

Find the cosine of the angle between the vectors $\mathbf{a}=2\mathbf{i}-3\mathbf{j}+2\mathbf{k}$ and $\mathbf{b}=\mathbf{i}+2\mathbf{j}-\mathbf{k}$ and determine if the angle is acute or obtuse.

Solution

$$\cos\theta=\frac{(2\mathbf{i}-3\mathbf{j}+2\mathbf{k})\cdot(\mathbf{i}+2\mathbf{j}-\mathbf{k})}{\sqrt{2^2+3^2+2^2}\sqrt{1^2+2^2+(-1)^2}}$$

$$\cos\theta=\frac{2-6-2}{\sqrt{17}\sqrt{6}}=-\sqrt{\frac{6}{17}}<0$$

θ is an obtuse angle

Properties of Scalar Product

(i) $\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$

(ii) $\mathbf{a}\cdot(\mathbf{b}\pm\mathbf{c})=(\mathbf{a}\cdot\mathbf{b})\pm(\mathbf{a}\cdot\mathbf{c})$

(iii) $\lambda(\mathbf{a}\cdot\mathbf{b})=(\lambda\mathbf{a})\cdot\mathbf{b}=\mathbf{a}\cdot(\lambda\mathbf{b})$

(iv) $\mathbf{a}\cdot\mathbf{a}=|\mathbf{a}|^2 \Leftrightarrow |\mathbf{a}|=\sqrt{\mathbf{a}\cdot\mathbf{a}}$ note the relationship between the scalar product and the modulus

Given that $|\mathbf{a}|=2,\ |\mathbf{b}|=3,\ \mathbf{a}\cdot\mathbf{b}=-1$,

(i)
$$\begin{aligned}&(\mathbf{a}-\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b})\\&=2\mathbf{a}\cdot\mathbf{a}+\mathbf{a}\cdot\mathbf{b}-2\mathbf{b}\cdot\mathbf{a}-\mathbf{b}\cdot\mathbf{b}\\&=2|\mathbf{a}|^2-\mathbf{a}\cdot\mathbf{b}-|\mathbf{b}|^2\\&=2(2^2)-(-1)-3^2\\&=0\end{aligned}$$

(ii)
$$\begin{aligned}&(2\mathbf{a}-\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b})\\&=4\mathbf{a}\cdot\mathbf{a}+2\mathbf{a}\cdot\mathbf{b}-2\mathbf{b}\cdot\mathbf{a}-\mathbf{b}\cdot\mathbf{b}\\&=4|\mathbf{a}|^2-|\mathbf{b}|^2\\&=4(2^2)-3^2\\&=7\end{aligned}$$

(iii)
$$\begin{aligned}|\mathbf{a}-2\mathbf{b}|&=\sqrt{(\mathbf{a}-2\mathbf{b})\cdot(\mathbf{a}-2\mathbf{b})}\\&=\sqrt{\mathbf{a}\cdot\mathbf{a}-2\mathbf{a}\cdot\mathbf{b}-2\mathbf{b}\cdot\mathbf{a}+4\mathbf{b}\cdot\mathbf{b}}\\&=\sqrt{|\mathbf{a}|^2-4\mathbf{a}\cdot\mathbf{b}+4|\mathbf{b}|^2}=\ldots=2\sqrt{11}\end{aligned}$$

Applications of Scalar Product

Points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, relative to the origin O. Given that P is the point on the line OB such that $\overrightarrow{AP} \perp \overrightarrow{OB}$, it can be shown that

$$\overrightarrow{OP} = \left(\frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{b}|}\right)\frac{\mathbf{b}}{|\mathbf{b}|} = \left(\mathbf{a}\cdot\hat{\mathbf{b}}\right)\hat{\mathbf{b}}$$

P is the foot of perpendicular from A to the line OB and P is also the point on the line OB nearest to A.
Thus, the **length of projection of vector a onto vector b** is given by $\left|\overrightarrow{OP}\right| = \left|\mathbf{a}\cdot\hat{\mathbf{b}}\right|$.

Points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, relative to the origin O.

Given that $\mathrm{a} = 2\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$ and $\mathrm{b} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, find the length of projection of **a onto vector b.**
Find the position vector of the foot of the perpendicular from A to OB.

Solution $\quad OP = \left|\mathbf{a}\cdot\hat{\mathbf{b}}\right| = \dfrac{\left|(2\mathbf{i}-3\mathbf{j}+2\mathbf{k})\cdot(\mathbf{i}+2\mathbf{j}-\mathbf{k})\right|}{\sqrt{1^2+2^2+(-1)^2}} = \dfrac{|2-6-2|}{\sqrt{6}} = \sqrt{6}$

$$\overrightarrow{OP} = \left(\mathbf{a}\cdot\hat{\mathbf{b}}\right)\hat{\mathbf{b}} = -\left(\frac{6}{\sqrt{6}}\right)\frac{\mathbf{i}+2\mathbf{j}-\mathbf{k}}{\sqrt{6}} = -\mathbf{i} - 2\mathbf{j} + \mathbf{k}$$

Vector (Cross) Product

Let $\mathbf{a}$ and $\mathbf{b}$ be two non-zero vectors that are represented by $\overrightarrow{OA}$ and $\overrightarrow{OB}$ respectively.

$$\mathbf{a}\times\mathbf{b} = \left(|\,\mathbf{a}\,||\,\mathbf{b}\,|\sin\theta\right)\hat{\mathbf{n}}$$

where $\theta = \angle AOB$ is the angle between $\mathbf{a}$ and $\mathbf{b}$, and $\hat{\mathbf{n}}$ is the unit vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

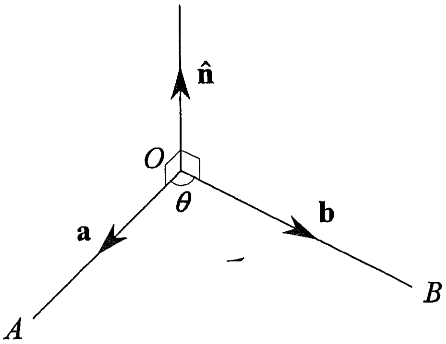


Figure 1

It follows that two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ are **perpendicular** if and only if $|\,\mathbf{a}\times\mathbf{b}\,| = |\,\mathbf{a}\,||\,\mathbf{b}\,|$.
In particular, $\mathbf{i}\times\mathbf{j} = \mathbf{k}$, $\mathbf{j}\times\mathbf{k} = \mathbf{i}$ and $\mathbf{k}\times\mathbf{i} = \mathbf{j}$.

Two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ are **parallel** if and only if $\mathbf{a}\times\mathbf{b} = \mathbf{0}$.
In particular, $\mathbf{i}\times\mathbf{i} = \mathbf{j}\times\mathbf{j} = \mathbf{k}\times\mathbf{k} = \mathbf{0}$.

Properties of Vector Product

1. $\mathbf{a}\times\mathbf{b} = -(\mathbf{b}\times\mathbf{a})$.
2. $|\,\mathbf{a}\times\mathbf{b}\,| = |\,\mathbf{b}\times\mathbf{a}\,| = |\,\mathbf{a}\,||\,\mathbf{b}\,||\sin\theta\,| \le |\,\mathbf{a}\,||\,\mathbf{b}\,|$.
3. $\mathbf{a}\times(\mathbf{b}\pm\mathbf{c}) = (\mathbf{a}\times\mathbf{b})\pm(\mathbf{a}\times\mathbf{c})$.
4. $\lambda(\mathbf{a}\times\mathbf{b}) = (\lambda\mathbf{a})\times\mathbf{b} = \mathbf{a}\times(\lambda\mathbf{b})$.
5. $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{a} = (\mathbf{a}\times\mathbf{b})\cdot\mathbf{b} = 0$ So $\mathbf{a}\times\mathbf{b}\perp\mathbf{a}$ and $\mathbf{a}\times\mathbf{b}\perp\mathbf{b}$.

Examples of how the properties are used

$$\begin{aligned}|(\mathbf{a}+2\mathbf{b})\times(3\mathbf{a}-\mathbf{b})| &= |3\mathbf{a}\times\mathbf{a}-\mathbf{a}\times\mathbf{b}+6\mathbf{b}\times\mathbf{a}-2\mathbf{b}\times\mathbf{b}| \\ &= |\mathbf{0}+\mathbf{b}\times\mathbf{a}+6\mathbf{b}\times\mathbf{a}+\mathbf{0}| \\ &= 7|\mathbf{b}\times\mathbf{a}|\end{aligned}$$

$$\begin{aligned}|(\alpha\mathbf{a}+\beta\mathbf{b})\times(\alpha\mathbf{a}-\beta\mathbf{b})| &= |\alpha^2\mathbf{a}\times\mathbf{a}-\alpha\beta\mathbf{a}\times\mathbf{b}+\alpha\beta\mathbf{b}\times\mathbf{a}-\beta^2\mathbf{b}\times\mathbf{b}| \\ &= |\alpha\beta||\mathbf{0}+\mathbf{b}\times\mathbf{a}+\mathbf{b}\times\mathbf{a}+\mathbf{0}| \qquad \left[\text{Note that:}|\alpha\mathbf{a}|=|\alpha||\mathbf{a}|\right] \\ &= 2|\alpha\beta||\mathbf{b}\times\mathbf{a}|\end{aligned}$$

Given that $(\mathbf{u}+\mathbf{v})\times(\mathbf{u}-\mathbf{v})=\mathbf{0}$, what can be deduced about the vectors $\mathbf{u}$ and $\mathbf{v}$?

Solution:

$$\begin{aligned}(\mathbf{u}+\mathbf{v})\times(\mathbf{u}-\mathbf{v}) &= \mathbf{0} \\ \mathbf{u}\times\mathbf{u}-\mathbf{u}\times\mathbf{v}+\mathbf{v}\times\mathbf{u}-\mathbf{v}\times\mathbf{v} &= \mathbf{0} \\ \mathbf{0}+2\mathbf{v}\times\mathbf{u}-\mathbf{0} &= \mathbf{0} \\ \mathbf{v}\times\mathbf{u} &= \mathbf{0}\end{aligned}$$

Then, $\mathbf{u}=\mathbf{0}$ or $\mathbf{v}=\mathbf{0}$ or $\mathbf{u}\parallel\mathbf{v}$

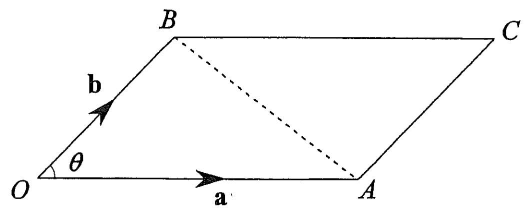
Let $A(a_1,a_2,a_3)$ and $B(b_1,b_2,b_3)$ be two points in three-dimensional space and let the position vectors of A and B with respect to the origin O be $\mathbf{a}$ and $\mathbf{b}$ respectively.

Then vector (cross) product $\mathbf{a}\times\mathbf{b}$, is the vector given by

$$\mathbf{a}\times\mathbf{b}=\begin{pmatrix}a_1\\a_2\\a_3\end{pmatrix}\times\begin{pmatrix}b_1\\b_2\\b_3\end{pmatrix}=\begin{pmatrix}a_2b_3-a_3b_2\\-(a_1b_3-a_3b_1)\\a_1b_2-a_2b_1\end{pmatrix}=\begin{pmatrix}a_2b_3-a_3b_2\\a_3b_1-a_1b_3\\a_1b_2-a_2b_1\end{pmatrix} \quad \text{(MF27)}$$

Applications of Vector Product

Let the points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ with respect to the origin O, and let θ be the angle between $\mathbf{a}$ and $\mathbf{b}$. Let C be the point such that $OACB$ is a parallelogram.



Then we have

(1) Area of triangle OAB is $\frac{1}{2}|\mathbf{a}\times\mathbf{b}|$.

(2) Area of parallelogram $OACB$ is $|\mathbf{a}\times\mathbf{b}|$.

(3) Distance from B to OA is $|\mathbf{b}\times\hat{\mathbf{a}}|$.

(Recall the **length of projection** of $\overrightarrow{OB}$ onto $\overrightarrow{OA}$ is $|\mathbf{b}\cdot\hat{\mathbf{a}}|$)

Revision Tutorial Questions

1 For each question, indicate the letter corresponding to the correct answer.

	Question	Answer
I	The modulus of the vector $6\mathbf{i}-2\mathbf{j}-3\mathbf{k}$ is **(a)** $\sqrt{23}$ **(b)** $\sqrt{11}$ **(c)** 1 **(d)** 49 **(e)** 7	
II	The unit vector in the direction of the vector $-\mathbf{i}+2\mathbf{j}-2\mathbf{k}$ is **(a)** $-\frac{1}{3}\mathbf{i}+\frac{2}{3}\mathbf{j}-\frac{2}{3}\mathbf{k}$ **(b)** $\frac{1}{3}\mathbf{i}-\frac{2}{3}\mathbf{j}+\frac{2}{3}\mathbf{k}$ **(c)** $\pm\left(\frac{1}{3}\mathbf{i}-\frac{2}{3}\mathbf{j}+\frac{2}{3}\mathbf{k}\right)$ **(d)** $-\frac{1}{5}\mathbf{i}+\frac{2}{5}\mathbf{j}-\frac{2}{5}\mathbf{k}$ **(e)** $-\frac{1}{9}\mathbf{i}+\frac{2}{9}\mathbf{j}-\frac{2}{9}\mathbf{k}$	
III	If $\mathbf{a}=2\mathbf{i}-\mathbf{j}$ and $\mathbf{b}=2\mathbf{i}+\mathbf{j}$ then $\mathbf{a}\cdot\mathbf{b}$ is **(a)** $4\mathbf{i}-\mathbf{j}$ **(b)** 3 **(c)** 0 **(d)** $4\mathbf{i}^2-\mathbf{j}^2$ **(e)** 5	
IV	If $\mathbf{a}=\mathbf{i}+\mathbf{j}$ and $\mathbf{b}=2\mathbf{i}-\mathbf{j}$ then $\mathbf{a}\times\mathbf{b}$ is **(a)** $2\mathbf{i}-\mathbf{j}$ **(b)** 0 **(c)** $3\mathbf{k}$ **(d)** $-3\mathbf{k}$ **(e)** 1	
V	It is given that the points *A*, *B* and *C* are collinear. Referred to the origin *O*, $\overrightarrow{OA}=\mathbf{i}+\mathbf{j}$, $\overrightarrow{OB}=2\mathbf{i}-\mathbf{j}+\mathbf{k}$ and $\overrightarrow{OC}=3\mathbf{i}+a\mathbf{j}+b\mathbf{k}$. Find *a* and *b*. **(a)** $a=-3, b=2$ **(b)** $a=3, b=-2$ **(c)** $a=0, b=1$ **(d)** $a=-1, b=0$ **(e)** $a=6, b=-1$	
VI	It is given that the points *A*, *B* and *C* are collinear. If $\mathbf{a}=-2\mathbf{b}+3\mathbf{c}$ then *C* divides *AB* in the ratio **(a)** 1:2 **(b)** 2:3 **(c)** 3:2 **(d)** -2:3 **(e)** 2:1	
VII	The angle between the vectors $\begin{pmatrix}-4\\3\end{pmatrix}$ and $\begin{pmatrix}1\\1\end{pmatrix}$ is given by **(a)** $\cos^{-1}(0)$ **(b)** $\cos^{-1}(1)$ **(c)** $\cos^{-1}\left(\frac{\sqrt{2}}{10}\right)$ **(d)** $\sin^{-1}\left(\frac{-1}{5\sqrt{2}}\right)$ **(e)** $\cos^{-1}\left(\frac{-1}{5\sqrt{2}}\right)$	
VIII	**a**, **b** and **c** are the position vectors of the vertices of a triangle. The area of the triangle can be expressed as **(a)** $(\mathbf{b}-\mathbf{a})\times(\mathbf{c}-\mathbf{a})$ **(b)** $\frac{1}{2}\lvert\mathbf{b}\times\mathbf{c}\rvert$ **(c)** $\frac{1}{2}\lvert(\mathbf{b}-\mathbf{a})\times(\mathbf{c}-\mathbf{b})\rvert$ **(d)** $\frac{1}{2}\lvert\mathbf{b}-\mathbf{a}\rvert\times\lvert\mathbf{c}-\mathbf{a}\rvert$ **(e)** $\lvert(\mathbf{b}-\mathbf{a})\cdot(\mathbf{c}-\mathbf{a})\rvert$	
IX	Given vectors **a** and **b**, and scalar λ, which statement is false in general. **(a)** $\mathbf{a}\times\mathbf{a}=\mathbf{0}$ **(b)** $\lvert\lambda\mathbf{a}\cdot\mathbf{b}\rvert=\lambda\lvert\mathbf{a}\cdot\mathbf{b}\rvert$ **(c)** $\mathbf{a}\cdot(\mathbf{b}\times\mathbf{a})=0$ **(d)** $(\mathbf{a}-\mathbf{b})\times(\mathbf{a}+\mathbf{b})=2\mathbf{a}\times\mathbf{b}$ **(e)** $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=\lvert\mathbf{a}\rvert^2-\lvert\mathbf{b}\rvert^2$	

X Referred to the origin O, the points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively.

Which of the following conditions below imply that $(\mathbf{a}-\mathbf{b})\times(\mathbf{a}+\mathbf{b})=\mathbf{0}$?

(1) $\mathbf{a}$ and $\mathbf{b}$ are perpendicular
(2) $\mathbf{a}$ and $\mathbf{b}$ are parallel
(3) Either $\mathbf{a}$ or $\mathbf{b}$ (or both) are the zero vector.
(4) The points O, A and B are collinear.
(5) $|\mathbf{a}|=|\mathbf{b}|$

(a) All of the above **(b)** (1) and (5) only **(c)** (2), (3) and (4) only
(d) (2) and (3) only **(e)** None of the above

Source: ACJC Prelim 9758/2018/01/Q3(b)

2 Referred to the origin O, points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, where $\mathbf{a}$ is a unit vector, $|\mathbf{b}|=3$, $|\mathbf{c}|=\sqrt{3}$ and angle AOC is $\frac{\pi}{6}$ radians. Given that $3\mathbf{a}+\mathbf{c}=k\mathbf{b}$, where $k\neq 0$, by considering $(3\mathbf{a}+\mathbf{c})\cdot(3\mathbf{a}+\mathbf{c})$, find the exact values of k. [4]

Source : PJC Prelim 9758/2018/02/Q1 (part)

3 Referred to the origin O, points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. The point D lies on OB, between O and B, such that $\overrightarrow{AD}$ is perpendicular to $\overrightarrow{OB}$. It is given that $|\mathbf{a}|=4$, $|\mathbf{b}|=8$ and $\angle AOB=60^\circ$.

Show that $\overrightarrow{OD}=\frac{1}{4}\mathbf{b}$. [2]

Source : ACJC Prelim 9758/2022/01/Q3 (part)

4 With reference to the origin O, the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, where points O, A, B and C are not collinear. It is given that $|\mathbf{a}|=\sqrt{10}$, $|\mathbf{b}|=20$, $\mathbf{a}\cdot\mathbf{b}=40$ and the point N has position vector $\dfrac{5\mathbf{a}+3\mathbf{b}}{8}$. Find the exact area of triangle ONB. [5]

Source: EJC Prelim 9758/2018/01/Q5

5 Referred to the origin O, the points P and Q have position vectors $\mathbf{p}$ and $\mathbf{q}$ where $\mathbf{p}$ and $\mathbf{q}$ are non-parallel, non-zero vectors. Point R is on PQ produced such that $PQ:QR=1:\lambda$. Point M is the mid-point of OR.

(i) Find the position vector of R in terms of λ, $\mathbf{p}$ and $\mathbf{q}$. [1]

F is a point on OQ such that F, P and M are collinear.

(ii) Find the ratio $OF:FQ$, in terms of λ. [4]

Source: MI Prelim 9758/2018/02/Q4 (modified)

6 Relative to the origin O, the position vectors of points A, B and C are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively. It is given that $\mathbf{a}$ and $\mathbf{b}-\mathbf{a}$ are perpendicular and C lies on AB produced such that $AC:AB = 4:3$.

(i) If $\mathbf{a}$ is a unit vector, show that $|\mathbf{b}| > 1$. [3]

Given further that $|\mathbf{b}| = 2$, find the angle between $\mathbf{a}$ and $\mathbf{b}$.

[1]

(ii) By expressing $\mathbf{c}$ in terms of $\mathbf{a}$ and $\mathbf{b}$, show that $|\mathbf{c}\cdot\mathbf{a}| = \mathbf{a}\cdot\mathbf{a}$. [3]

Hence state the length of projection of $\mathbf{c}$ onto $\mathbf{a}$ in terms of $\mathbf{a}$. [1]

(iii) Give a geometrical interpretation of $|\mathbf{c}\times\mathbf{a}|$ and hence evaluate $\dfrac{|\mathbf{c}\times\mathbf{a}|}{|\mathbf{b}\times\mathbf{a}|}$. [2]

Source: SAJC Prelim 9758/2018/01/Q6

7 Relative to the origin O, the position vectors of points A and B are $\mathbf{a}$ and $\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-zero and non-parallel vectors.
The point C with position vector $\mathbf{c}$ lies on the line segment AB such that $AC : CB$ is $\lambda : 1-\lambda$. It is given that $\mathbf{b}$ is a unit vector, $|\mathbf{a}| = \dfrac{4}{3}$ and the angle formed between OA and OB is $120°$.

(i) Find the value of λ such that the points O, A and C form a right angle AOC. [4]

(ii) Find $\mathbf{m}$, the position vector of M, the midpoint of AC, in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]

A circle is drawn with AC as its diameter and O is a point on the circumference of the circle drawn.

(iii) Determine if OB is a tangent to the circle described above. [3]

(iv) Give a geometrical interpretation of $|\mathbf{b}\bullet\mathbf{m}|$. Hence, explain $(\mathbf{b}\bullet\mathbf{m})\mathbf{b}$ in terms of its magnitude and direction. [2]

Source: ASRJC Prelim 9758/2022/02/Q2

8 Relative to the origin O, the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively. It is given that λ and μ are non-zero numbers such that $\lambda\mathbf{a} + \mu\mathbf{b} - \mathbf{c} = \mathbf{0}$ and $\lambda + \mu = 1$.

(i) Show that the points A, B and C are collinear. [3]

The angle between $\mathbf{a}$ and $\mathbf{b}$ is known to be obtuse and that $|\mathbf{a}| = 2$.

(ii) If k denotes the area of triangle OAB, show that $(\mathbf{a}\cdot\mathbf{b})^2 = 4\left(|\mathbf{b}|^2 - k^2\right)$. [3]

D is a point on the line segment AB with position vector $\mathbf{d}$.

(iii) It is given that area of triangle OAB is 6 units^2, $|\mathbf{b}| = 10$ and that AOD is 90^0.

By finding the value of $\mathbf{a}\cdot\mathbf{b}$, find $\mathbf{d}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [4]

Source: CJC Prelim 9758/2022/02/Q4

9 **(a)** Referred to the origin O, the fixed points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-zero vectors. It is given that O divides the line segment AC in the ratio 1:3. Point D divides the line segment AB in the ratio of $\lambda : 1-\lambda$. Point E divides the line segment BC in the ratio $\mu : 1-\mu$.

Show that the area of triangle ODE can be written as $k|\mathbf{a}\times\mathbf{b}|$, where k is a constant to be found in terms of μ and λ. [5]

(b) A variable point S has position vector $\mathbf{s}$ relative to the origin O. Given that $\mathbf{s}\times\mathbf{b} = -\mathbf{b}\times\mathbf{a}$, describe geometrically the set of all possible positions of the point S. [3]

Source: DHS Prelim 9758/2022/02/Q4

10 Relative to the origin O, the position vectors of the points P, Q and R are $\mathbf{i}$, $2\mathbf{j} - t\mathbf{k}$ and $t\mathbf{k}$ respectively, where t is a fixed constant. The points A and B divides both line segments PQ and QR respectively in the same ratio of $\mu : 1-\mu$, where μ is a parameter such that $0 < \mu < 1$.

(a) Find the vector $\overrightarrow{AB}$ in terms of t and μ. [3]

(b) Determine whether the points O, A, B are collinear. [1]

(c) Find the values of μ such that the length of projection of $\overrightarrow{AB}$ onto $\begin{pmatrix}3\\4\\0\end{pmatrix}$ is $\frac{1}{5}$ unit. [2]

(d) Given that angle AOB is a right angle, find the set of possible values of t, justifying your answer clearly. [3]

Soucee: JPJC Prelim 9758/2022/02/Q2

11 **(a)** With reference to the origin O, the points A, B and X are $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$ and $\overrightarrow{OX} = \frac{1}{8}\mathbf{a} + \frac{3}{8}\mathbf{b}$. The point Y lies on AB such that O, X and Y are collinear.

Express $\overrightarrow{OY}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ and find the ratio of $AY : YB$. [5]

(b) The points P, Q and R have position vectors $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$ respectively. P and Q are fixed and R varies. Describe geometrically the set of possible positions of the point R such that

(i) $(\mathbf{r}-\mathbf{p})\times\mathbf{q} = \mathbf{0}$, [2]

(ii) $(\mathbf{r}-\mathbf{p})\cdot\mathbf{q} = 0$. [2]